



ENT164 Introduction to Making

Course Syllabus for Fall 2026

Course Information

Credits: 3

Meeting Time: Thursdays 1:20-4:20pm

Location: JCC 301 / Nolop

Instructor & TAs

Name	Role	Email	Office Hours
Thomas Van de Velde	Instructor	thomas.van_de_velde@tufts.edu	go.tufts.edu/meet_tvd_y
Alexander Beattie	TA	alexander.beattie@tufts.edu	By appointment
Patrice Pinardo	TA	patrice.pinardo@tufts.edu	By appointment

General Information

Overview

How can we harness maker tools and technology to build working prototypes that address real-world challenges?

This entrepreneurial course demystifies technology by teaching you to build smart device prototypes using physical making, basic electronics, and programming. By creating working prototypes, you'll master the iterative innovation process essential to entrepreneurship: build, test, learn, and improve. This practical approach transforms technology from an intimidating black box into an accessible tool for bringing your ideas to life. You'll develop an entrepreneurial mindset by learning to work as a team, fail fast, and iterate rapidly.

No Restrictions on Enrollment: All undergraduate and graduate students are welcome, including first-year undergrads. No prior experience in design thinking, innovation or programming is needed to take this class.

This class introduces basic programming techniques to make devices interact with their environment. Prior programming experience is not a requirement. Students will learn how to leverage AI to program microcontrollers so a general interest in programming is suggested.

Learning Outcomes

By the end of this course, you will be able to:

Make & Fabricate

- Design and fabricate physical objects using CAD, laser cutters, and 3D printers
- Build functional circuits and program microcontrollers using MicroPython and AI

Connect & Integrate

- Create smart IoT devices that sense, visualize, and respond to environmental data
- Integrate multiple technologies into a cohesive working prototype

Collaborate & Iterate

- Work effectively in multidisciplinary teams to design and build prototypes
- Apply the "build-measure-learn" loop through rapid iteration and failing fast

Communicate & Present

- Present working prototypes to peers and public audiences
- Articulate design decisions and lessons learned

Format of Classes and Assignments

This course is intended to be taught in-person. We expect students to make every effort to join us in the (physical) classroom unless they encounter extenuating circumstances. Important maker, prototyping, and programming skills will be demonstrated in class. If you need to miss class, you will need to make an appointment with an instructor or a TA to help you catch up.

For most weeks, our class time will be divided into two parts. We meet in the classroom for the first 60-90 minutes to recap the previous class, introduce foundational maker skills, and complete short exercises. We then move to Nolop for the second half of class to apply those maker skills and work on team projects.

Weekly Reflections

Reflection is a critical part of the learning process; research shows that we learn more effectively when we pause to think about what we've experienced. At the end of each class, you'll take a few minutes to reflect on what you learned, what challenged you, and how you might apply new skills to your projects. These brief reflections help solidify your learning and build self-awareness as a maker.

Expectation: Submit your weekly reflection within 24 hours of each class. Reflections are part of your Participation grade.

Course Modules

This course is divided into two modules:

Module 1: Foundational Skills. Learn to make physical things using hand making techniques, CAD tools, 3D printers, and laser cutting. You'll work on basic electronics projects to enable your projects to interact with the world. In Week 2, you'll be placed into teams of 4-5 students and collaborate on the Robo Challenge: design and build a robot that can move as many balls as possible between two lines. You'll also start defining the final project you'll present on demo day.

Module 2: Final Project Work. Build on the first module's foundation to develop your team's final project. You'll learn advanced skills including data sensing, visualization, and AI integration. Your team chooses the real-world challenge you want to tackle. Your final project must demonstrate the core maker techniques: laser cutting, 3D printing, electronics, connectivity, and AI. The module ends with a public demo day.

Norms and Expectations

Attendance Policy

- On-time, in-person class attendance and respectful class participation is required.
- If you need to miss class, arrive late, or leave early (e.g., to attend a job interview), or if you need to attend remotely due to extenuating circumstances (e.g., illness, travel overseas for family reasons), notify instructors by 5:30pm the day before class. You may also use the Short-Term Illness Notification tool in SIS.
- **Absences due to illness:** If you are too ill to attend class, please inform me as soon as possible. You do not need to provide medical documentation of any kind. Please do not come to class if you are sick; we will work with you to catch up on missed material.
- When doing group work, observe team process norms including equal participation of all team members. These norms will be introduced in the beginning of the semester.
- Students are expected to adhere to the school's honor code.
- Diversity is celebrated at Tufts, as it should be around the world. Diversity is multidimensional and spans gender identity, sexual orientation, race, prior experiences and more. Respectful, inclusive and equitable treatment of all students and instructors is a fundamental norm in this class.

Resources for Students

Religious Accommodations: Tufts University faculty, staff, and administration highly value and acknowledge the religious diversity of its student body. Students seeking religious accommodations related to their holy days are encouraged to collaborate with faculty to make arrangements during the first week of each semester. Consult the [Multifaith Calendar](#) for upcoming holidays, links to the University Religious Accommodations Policy, and members of the University Chaplaincy who are available to respond to questions on religious observances.

Accommodations for Students with Disabilities: Tufts is committed to providing equal access and support to all qualified students through the provision of reasonable accommodations. If you have a disability that requires reasonable accommodations, contact the StAAR Center at StaarCenter@tufts.edu or [617-627-4539](tel:617-627-4539). Please be aware that accommodations cannot be enacted retroactively, making timeliness a critical aspect for their provision.

Academic Support at the StAAR Center: The StAAR Center offers a variety of FREE resources to all students. Students may make an appointment to work on any writing-related project or assignment, attend subject tutoring in a variety of disciplines, or meet with an academic coach to hone skills like time management and navigating procrastination. Students can make an appointment for any of these services by visiting <https://students.tufts.edu/staar-center>.

Student Support, including Mental Health: As a student, there may be times when personal stressors or difficulties interfere with your academic performance or well-being. The Dean of Students Office offers support and care to undergraduates and graduate students who are experiencing difficulties, and can also aid faculty in their work with students. In addition, through Tufts' Counseling and Mental Health Service (CMHS) students can access mental health support 24/7, and they can provide information on additional resources. CMHS also provides confidential consultation, brief counseling, and urgent care at no cost for all Tufts undergraduates as well as for graduate students who have paid the student health fee. To make an appointment, call 617-627-3360. Please visit the CMHS website: <http://go.tufts.edu/Counseling> to learn more about their services and resources.

Ears for Peers: Tufts' anonymous and confidential student-run hotline. Open every night 7pm-7am. Call: 617-627-3888. Text: ears4peers.herokuapp

Academic Integrity

Tufts holds its students strictly accountable for adherence to academic integrity. The consequences for violations can be severe. It is critical that you understand the requirements of ethical behavior and academic work as described in Tufts' Academic Misconduct policy. If you ever have a question about the expectations concerning a particular assignment or project in this course, be sure to ask one of the instructors for clarification.

While collaboration is encouraged, individual assignments are expected to be submitted individually using the components provided to the student. Allocate sufficient time to request replacement components if required. It is your responsibility to pick up new components promptly when missing a class. Submitting an individual assignment as shared work will result in points being divided among the participants.

On the Use of AI

This course integrates AI as a core tool in the maker's workflow. You will learn to use AI to design, iterate, and build more advanced prototypes than would otherwise be possible. **AI use is encouraged and expected.** You may use tools like Claude, ChatGPT, Gemini, and other AI assistants for:

- Generating and debugging code for microcontrollers
- Brainstorming design ideas
- Understanding technical concepts
- Troubleshooting electronics and fabrication issues

Attribution is not required, but documenting your AI-assisted workflow is encouraged as a learning practice.

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Grading

Overview

Component	Weight
Individual Assignments	35%
Team assignments, presentations and reports (team grades are shared by default but may be adjusted based on individual performance)	35%
Peer and Self Evaluations (mid-term and final)	20%
Reflections and Participation	10%

Peer and Self Evaluations: Twice during the semester (mid-term and final), you'll assess your own contributions and those of your teammates. These evaluations help ensure fair grading for team projects and encourage accountability within your team. Your feedback is confidential and will be used to adjust individual grades when contributions are unequal.

Assignment Policies

- All assignments are submitted through **Canvas**.
- Assignments are typically due on Tuesdays by noon the day of class.
- There will be a 25% late fee for submissions received during the grace period (within 1 week of the submission deadline). We will not accept submissions beyond the grace period.
- **Missed work due to illness:** If you are sick and unable to complete an assignment on time, please contact me before the deadline to discuss an extension.
- **Missed presentations:** If you are too ill to present in class, we can arrange for you to present remotely or reschedule.

Course Materials

Materials: All materials are provided by the course at no cost to students. You will receive an electronics starter kit at the beginning of the semester. Additional materials and tools will be available at Nolop. No textbooks are required.

What to bring to class:

- Laptop (required)
- USB-C adapter if your laptop only has USB-A ports
- External mouse (recommended for CAD work)
- Your electronics kit (bring to every class)

Online platforms and tools:

- **Canvas** for assignment submissions, course materials, announcements, and updates
- **Claude/Gemini/ChatGPT** for code generation and electronics
- **Onshape** for 2D and 3D modelling
- **Thonny** for MicroPython programming

Course Schedule

Week	Date	Topic	Learning Outcomes	Individual	Team
1	9/10/26	Kickoff and Nolop Tour Welcome to the course! Meet your classmates and instructors, explore the makerspace, and get your first taste of hands-on making.	- Identify key equipment and safety protocols in the makerspace - Complete a simple hands-on build from start to finish		
2	9/17/26	Ideation and Prototyping Learn foundational techniques for turning ideas into tangible prototypes using sketching and physical mockups.	- Understand the Robo Challenge requirements and form your team - Rapidly sketch 3+ concept variations for your design - Build a foam core mockup to test form and mechanics	Onboarding Survey: Share your background and goals for the course	
3	9/24/26	From Physical to Digital: 2D Transition from hand-making to digital fabrication by designing in CAD and cutting with a laser.	- Create a laser-ready 2D design file from a sketch - Laser cut structural components		Team Agreement: Define roles, norms, and communication plan
4	10/1/26	From Physical to Digital: 3D Expand your digital fabrication skills to three dimensions with CAD modeling and 3D printing.	- Model 3D parts (mounts, brackets, custom components) 3D print functional parts for your project	Laser Cut Project: Present a functional laser-cut creation to the class	
5	10/8/26	Microcontrollers Bring your projects to life with electronics and learn to program microcontrollers using AI-assisted coding.	- Wire motors, sensors, and controls on a breadboard - Program basic movements using MicroPython and AI	3D Print Project: Present a functional 3D-printed part to the class	
6	10/15/26	Connectivity Make your devices communicate wirelessly and enable remote monitoring and control.	- Connect a microcontroller to a wireless network - Build a remote control interface	Electronics Project: Present a working circuit to the class	
7	10/22/26	Robo Challenge Making Time Apply everything you've learned to build a working robot with your team.	- Integrate all components into a fully functional robot - Debug and iterate based on testing results		
8	10/29/26	Robo Challenge Put your robot to the test in a competitive race and reflect on your team's journey.	- Demonstrate a working robot under real-world conditions - Identify what worked, what didn't, and what you'd do differently		Robot Race: Compete and demonstrate your team's robot

Course Schedule (continued)

Week	Date	Topic	Learning Outcomes	Individual	Team
9	11/5/26	Final Project Kickoff Begin your capstone project by brainstorming ideas and presenting initial concepts to the class.	• Generate and validate a product concept with your team • Pitch your idea and incorporate peer feedback	Peer Review: Provide constructive feedback on classmates' ideas; Self Review: Reflect on your Robo Challenge contributions	
10	11/12/26	Sensing and Visualization Learn to gather data from the physical world and present it in meaningful visual formats.	• Collect environmental data using sensors • Display sensor data in a real-time visualization • Apply sensing techniques to your final project		Project Checkpoint 1: Demo core functionality and technical approach
11	11/19/26	Making Devices Smart with AI Add intelligence to your hardware projects by integrating AI capabilities.	• Integrate AI-powered decision making into a physical device • Identify opportunities to enhance your final project with AI		Project Checkpoint 2: Demo integrated features and user experience
	11/26/26	THANKSGIVING - NO CLASS			
12	12/3/26	Final Project Making Time Dedicated workshop time to complete and polish your final project before the showcase.	• Complete a functional prototype ready for demonstration • Prepare a compelling presentation of your product		
13	12/10/26	Final Project Showcase Celebrate your accomplishments by presenting your finished product to the Tufts community.	• Present and demonstrate your finished product to an audience • Reflect on your growth and lessons learned throughout the course	Peer Evaluation: Assess teammates' contributions to the final project; Self Review: Reflect on your growth throughout the course	Final Product Demo: Present a working prototype to a public audience

This syllabus is subject to change. Students will be notified of any changes via Canvas.